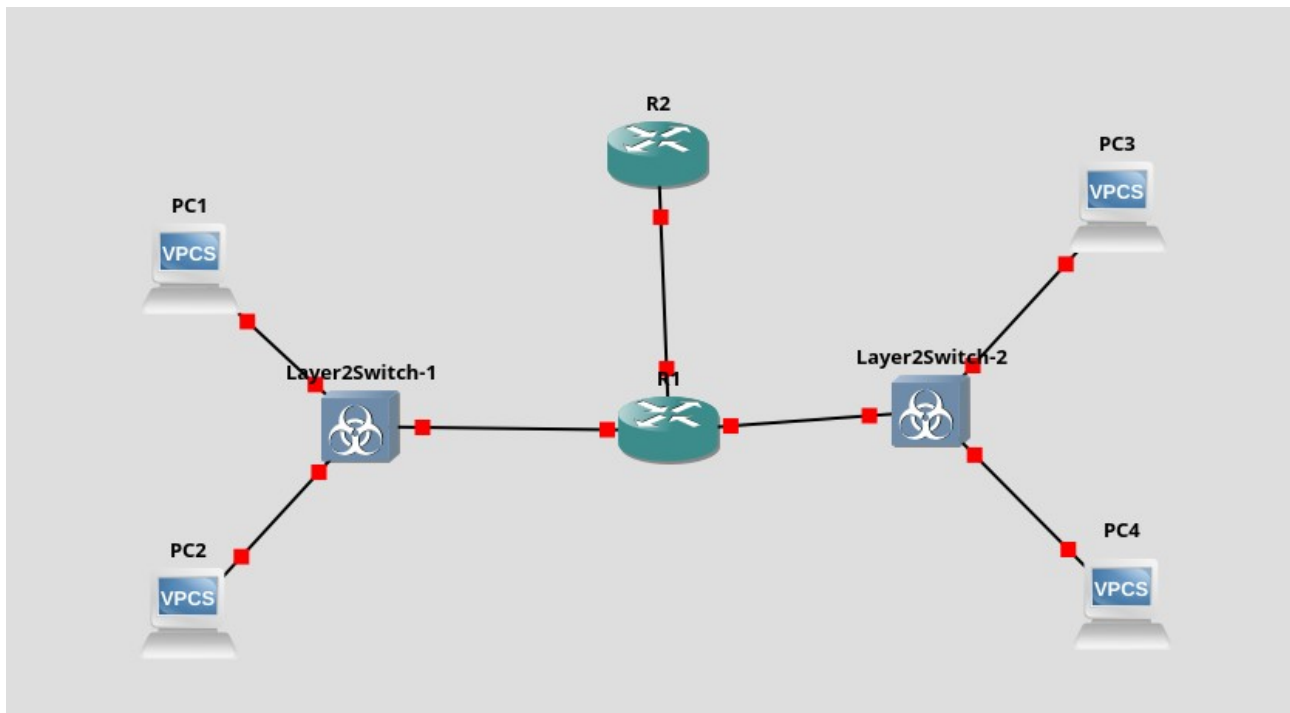


Module4 lab4

Корначенко Данил

1)



R1

R1#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R1(config)#interface FastEthernet1/0

R1(config-if)#ip address 192.168.1.1 255.255.255.0

R1(config-if)#no shutdown

R1(config-if)#exit

R1(config)#interface Ethernet2/0

R1(config-if)#ip address 192.168.2.1 255.255.255.0

R1(config-if)#no shutdown

R1(config-if)#exit

R1(config)#interface FastEthernet0/0

R1(config-if)#ip address 10.0.0.1 255.255.255.252

R1(config-if)#no shutdown

R1(config-if)#exit

R2

R2#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2(config)#interface FastEthernet0/0

R2(config-if)#ip address 10.0.0.2 255.255.255.252

R2(config-if)#no shutdown

R2(config-if)#exit

2) Настройка dhcp

Надо исключить статические ip и настроить пул адресов для LAN'ов

R2

```
R2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#ip dhcp excluded-address 192.168.1.1 192.168.1.10
R2(config)#ip dhcp excluded-address 192.168.2.1 192.168.2.10
R2(config)#ip dhcp pool LAN1
R2(dhcp-config)#network 192.168.1.0 255.255.255.0
R2(dhcp-config)#default-router 192.168.1.1
R2(dhcp-config)#dns-server 8.8.8.8
R2(dhcp-config)#exit
R2(config)#ip dhcp pool LAN2
R2(dhcp-config)#network 192.168.2.0 255.255.255.0
R2(dhcp-config)#default-router 192.168.2.1
R2(dhcp-config)#dns-server 8.8.8.8
R2(dhcp-config)#exit
R2(config)#end
R2#write
*Mar 1 00:35:32.455: %SYS-5-CONFIG_I: Configured from console by console
R2#write memory
Building configuration...
[OK]
```

Поскольку R2 находится в LAN3, нужно настроить ретрансляцию dhcp-запросов на R1

R1

```
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface FastEthernet1/0
R1(config-if)#ip helper-address 10.0.0.2
R1(config-if)#exit
R1(config)#interface Ethernet2/0
R1(config-if)#ip helper-address 10.0.0.2
R1(config-if)#exit
R1(config)#end
R1#wri
*Mar 1 00:44:20.323: %SYS-5-CONFIG_I: Configured from console by console
R1#write memory
Building configuration...
[OK]
```

3) Настраиваем маршруты

R2#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1

R2(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.1

R2(config)#end

R2#wri

*Mar 1 00:48:59.395: %SYS-5-CONFIG_I: Configured from console by console

R2#write memory

Building configuration...

[OK]

R2#show ip route

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/30 is subnetted, 1 subnets

C 10.0.0.0 is directly connected, FastEthernet0/0

S 192.168.1.0/24 [1/0] via 10.0.0.1

S 192.168.2.0/24 [1/0] via 10.0.0.1

Проверка

R2#ping 10.0.0.1 Type escape sequence to abort. Sending 5, 100-byte ICMP Echos to 10.0.0.1, timeout is 2 seconds: !!!! Success rate is 100 percent (5/5), round-trip min/avg/max = 60/60/64 ms	R1#ping 10.0.0.2 Type escape sequence to abort. Sending 5, 100-byte ICMP Echos to 10.0.0.2, timeout is 2 seconds: !!!! Success rate is 100 percent (5/5), round-trip min/avg/max = 60/74/92 ms
---	--

4) Получаем адреса и пингуемся

```
PC1> ip dhcp
DDORA IP 192.168.1.12/24 GW 192.168.1.1

PC1> show ip
NAME       : PC1[1]
IP/MASK    : 192.168.1.12/24
GATEWAY    : 192.168.1.1
DNS        : 8.8.8.8
DHCP SERVER : 10.0.0.2
DHCP LEASE : 86394, 86400/43200/75600
MAC        : 00:50:79:66:68:00
LPORT      : 22422
RHOST:PORT : 127.0.0.1:22423
MTU        : 1500

PC1>

PC2> ip dhcp
DDORA IP 192.168.1.13/24 GW 192.168.1.1

PC2> show ip
NAME       : PC2[1]
IP/MASK    : 192.168.1.13/24
GATEWAY    : 192.168.1.1
DNS        : 8.8.8.8
DHCP SERVER : 10.0.0.2
DHCP LEASE : 86392, 86400/43200/75600
MAC        : 00:50:79:66:68:01
LPORT      : 22424
RHOST:PORT : 127.0.0.1:22425
MTU        : 1500

PC2>

PC3> ip dhcp
DDORA IP 192.168.2.11/24 GW 192.168.2.1

PC3> show ip
NAME       : PC3[1]
IP/MASK    : 192.168.2.11/24
GATEWAY    : 192.168.2.1
DNS        : 8.8.8.8
DHCP SERVER : 10.0.0.2
DHCP LEASE : 86394, 86400/43200/75600
MAC        : 00:50:79:66:68:02
LPORT      : 22442
RHOST:PORT : 127.0.0.1:22443
MTU        : 1500

PC3>

PC4> ip dhcp
DDORA IP 192.168.2.12/24 GW 192.168.2.1

PC4> show ip
NAME       : PC4[1]
IP/MASK    : 192.168.2.12/24
GATEWAY    : 192.168.2.1
DNS        : 8.8.8.8
DHCP SERVER : 10.0.0.2
DHCP LEASE : 86396, 86400/43200/75600
MAC        : 00:50:79:66:68:03
LPORT      : 22444
RHOST:PORT : 127.0.0.1:22445
MTU        : 1500

PC4>
```

```
PC1> ping 192.168.1.13
84 bytes from 192.168.1.13 icmp_seq=1 ttl=64 time=8.093 ms
84 bytes from 192.168.1.13 icmp_seq=2 ttl=64 time=3.386 ms
84 bytes from 192.168.1.13 icmp_seq=3 ttl=64 time=1.340 ms
84 bytes from 192.168.1.13 icmp_seq=4 ttl=64 time=1.965 ms
84 bytes from 192.168.1.13 icmp_seq=5 ttl=64 time=1.911 ms

PC1>

PC2> ping 192.168.2.11
84 bytes from 192.168.2.11 icmp_seq=1 ttl=63 time=30.058 ms
84 bytes from 192.168.2.11 icmp_seq=2 ttl=63 time=18.004 ms
84 bytes from 192.168.2.11 icmp_seq=3 ttl=63 time=28.073 ms
84 bytes from 192.168.2.11 icmp_seq=4 ttl=63 time=20.748 ms
84 bytes from 192.168.2.11 icmp_seq=5 ttl=63 time=18.048 ms

PC2>

PC3> ping 192.168.2.12
84 bytes from 192.168.2.12 icmp_seq=1 ttl=64 time=11.319 ms
84 bytes from 192.168.2.12 icmp_seq=2 ttl=64 time=0.105 ms
84 bytes from 192.168.2.12 icmp_seq=3 ttl=64 time=1.208 ms
84 bytes from 192.168.2.12 icmp_seq=4 ttl=64 time=1.528 ms
84 bytes from 192.168.2.12 icmp_seq=5 ttl=64 time=0.087 ms

PC3>

PC4> ping 192.168.1.12
84 bytes from 192.168.1.12 icmp_seq=1 ttl=63 time=29.131 ms
84 bytes from 192.168.1.12 icmp_seq=2 ttl=63 time=19.073 ms
84 bytes from 192.168.1.12 icmp_seq=3 ttl=63 time=18.750 ms
84 bytes from 192.168.1.12 icmp_seq=4 ttl=63 time=18.161 ms
84 bytes from 192.168.1.12 icmp_seq=5 ttl=63 time=18.650 ms

PC4>
```

5) Перехват трафика

No.	Time	Delta	Source	Destination	Protocol	Length	Info
71	71.950924	0.014946	192.168.1.1	192.168.1.12	DHCP	342	DHCP ACK - Transaction ID 0x3f25dd47
56	57.612794	0.029801	192.168.1.1	192.168.1.12	DHCP	342	DHCP ACK - Transaction ID 0xab8d6e05
68	70.935876	13.323082	0.0.0.0	255.255.255.255	DHCP	406	DHCP Discover - Transaction ID 0x3f25dd47
51	56.582958		0.0.0.0	255.255.255.255	DHCP	406	DHCP Discover - Transaction ID 0xab8d6e05
69	70.949993	0.014117	192.168.1.1	192.168.1.12	DHCP	342	DHCP Offer - Transaction ID 0x3f25dd47
52	56.611283	0.028325	192.168.1.1	192.168.1.12	DHCP	342	DHCP Offer - Transaction ID 0xab8d6e05
70	71.935978	0.985985	0.0.0.0	255.255.255.255	DHCP	406	DHCP Request - Transaction ID 0x3f25dd47
55	57.582993	0.971710	0.0.0.0	255.255.255.255	DHCP	406	DHCP Request - Transaction ID 0xab8d6e05

Discover — Клиент отправляет широковещательный запрос в поиске DHCP

Offer — Unicast между роутерами, затем R1 ретранслирует его в LAN1. Маршрутизатор получив discover подставил свой адрес и так R2 понял из какого пула адресов нужно выдать адрес.

Request — Клиент широковещательно говорит, что принял адрес от сервера.

Ack — Сервер подтверждает выдачу адреса

6) Сохраняем конфигурации устройств

```
ls
R1.cfg R2.cfg

~/Projects/eltex_summer_course/module4/lab4
```